ElectricRV Dual-Fuel Smart Thermostat — User Manual

Model: DT-1 (Dettson/Gree edition) **Manufacturer:** ElectricRV **Document:** User Manual (for the homeowner)

DRAFT v0.1 — PRE-RELEASE, NOT FOR FIELD USE

This manual describes a product that is still in development. Some behaviour described here has not yet been verified on installed equipment. Sections that depend on pending verification are clearly marked. Do not rely on this document to operate a live heating system.

Product and model names are provisional branding and may change before release.

Revision history

Revision	Date	Changes
0.1	2026-06-11	Initial draft

About this manual

This manual is for the people who live with the thermostat: how to read the screen, set temperatures, choose modes, use schedules and remote sensors, and understand what the alerts mean. It assumes the thermostat has already been installed and commissioned.

If you are installing, wiring, or configuring the thermostat for the first time, use the separate **Installation & Commissioning Manual**. Nothing in this user manual involves opening the wall unit, the furnace, or the heat pump.

Copyright and disclaimer

This thermostat is an experimental, open-source controller for a gas furnace and heat pump. It is **not** a certified thermostat. The furnace and heat pump keep all of their own built-in safety systems — the thermostat only *requests* heating or cooling and the equipment decides whether it is safe to run — but installing a non-certified control on a gas appliance can affect your equipment warranty, may be regulated work in your area, and could affect your home insurance. Discuss this with your installer.

ElectricRV provides this product and documentation “as is”, without warranty of any kind. Operating a gas appliance always carries risk: keep a working carbon monoxide (CO) alarm in your home, and read the safety section of this manual before using the system.

Contents

Welcome

Your ElectricRV Dual-Fuel Smart Thermostat controls two pieces of heating and cooling equipment as one system:

- a **Dettson Chinook modulating gas furnace**, and
- a **Gree-built heat pump**, which provides both heating and air conditioning.

“Dual fuel” means the thermostat automatically chooses the best heat source for the conditions. In mild weather it favours the heat pump, which moves heat instead of burning fuel. In cold weather, when a heat pump becomes less effective, it switches to gas heat. You set the temperature; the thermostat decides how to get there.

What makes it different

- **True modulating gas control.** Most thermostats can only switch a furnace on or off. This thermostat speaks the furnace’s own digital language and asks for exactly the amount of heat needed — anywhere from 40 % to 100 % of the furnace’s capacity. The result is longer, gentler, quieter heating cycles and steadier room temperatures, instead of blasts of full heat followed by cool-downs.
- **Automatic dual-fuel changeover.** The thermostat watches the outdoor temperature and switches between heat pump and gas heat around a configurable outdoor “balance point”, with built-in protection so the switch never harms the compressor.
- **Heating and cooling in one.** Modes for heat, cool, automatic changeover between them, and a gas-only emergency heat mode.
- **A real touchscreen on the wall.** A 4.3-inch colour touchscreen shows the current temperature, both setpoints, what equipment is running, the outdoor temperature, and any alerts — and it keeps working even if your home network is down.
- **Remote room sensors.** Add small wireless temperature and occupancy sensors in the rooms you actually use. The thermostat blends their readings and gives more weight to occupied rooms (“follow me” comfort).
- **Local-first smart control.** The thermostat connects to **Home Assistant**, a free home-automation platform that runs on your own equipment in your own home. Schedules, phone control, history graphs, and notifications all work without any

cloud service, subscription, or account. If Home Assistant or your network goes down, the thermostat keeps controlling temperature on its own.

- **Safety-first design.** If anything goes wrong — a sensor fails, the controller locks up, the network disappears — the thermostat's designed response is always to *stop asking for heating or cooling*, never to run equipment blindly. The furnace's and heat pump's own built-in protections remain fully in charge of their machinery at all times.

What you need

Item	Required?
Dettson Chinook furnace + Gree-built heat pump (professionally installed)	Yes
The DT-1 wall thermostat (professionally installed and commissioned)	Yes
Home Assistant on your home network	For schedules, phone app, and remote sensors
Remote room sensors (Zigbee or ESPHome-based)	Optional
A working carbon monoxide (CO) alarm	**Yes — see the safety section**

You can use the thermostat entirely from the wall screen if you wish. Home Assistant adds scheduling, the phone app, and remote sensors, but day-to-day temperature control never depends on it.

Safety information for users

Your thermostat controls a **gas appliance** and a **heat pump compressor**. The thermostat never operates the gas valve or the compressor directly — it sends requests, and the furnace's certified control board and the heat pump's own electronics decide whether and how to run, with all of their built-in safety systems (flame sensing, overheat limits, pressure switches, compressor protections) always active. Even so, please read and follow this section.

Carbon monoxide (CO) alarm — required

Install and maintain a working CO alarm in your home. Any home with a gas-burning appliance should have one regardless of which thermostat is on the wall. Test it monthly and replace batteries per the alarm manufacturer's instructions.

If your CO alarm sounds: get everyone outside to fresh air immediately, then call emergency services or your gas utility from outside. Do not re-enter until told it is safe.

If you smell gas: do not touch any electrical switches (including the thermostat). Leave the building immediately and call your gas utility from outside.

Rules for safe use

- **Never block or cover** the furnace's air intakes, exhaust vents, supply registers, or return grilles, and keep the area around the outdoor heat pump unit clear of snow, leaves, and debris.
- **Do not open** the thermostat, the furnace cabinet, or the heat pump. There are no user-serviceable parts inside any of them.
- **Do not press the thermostat's screen with sharp objects** or mount anything over it.
- The thermostat is powered from the furnace. **Do not switch off the furnace's power** to "reset" the system unless instructed by a service technician — in winter, a powered-down system cannot protect your home from freezing.

Understanding safety-related alerts

The thermostat shows alerts in a banner on the home screen (and, if Home Assistant is connected, as notifications on your phone). The full alert table is in the *Alerts and troubleshooting* section. The safety-relevant ones, in plain language:

Alert	What it means	What to do
Furnace fault (with a code)	The furnace's own control board has reported a problem and may have shut itself down. The thermostat displays the code; it cannot override the furnace.	Note the code and call your HVAC service company.
Heating stopped — no progress	Gas heat ran continuously for 4 hours without reaching the target, so the thermostat stopped the call and raised an alert. This can mean an open window or door, an undersized setting on an extreme day, or an equipment problem.	Check for open windows/doors and a clogged filter. If it recurs, call a professional.
No heat — system in safe state	The thermostat detected a fault it could not resolve and has stopped all requests. In winter this is itself urgent: an unheated home risks frozen pipes. The alert escalates the longer it persists.	Call your installer or HVAC service company promptly. Ask your installer about the backup thermostat changeover they prepared at installation.
Backup sensor mode	All room sensors became unavailable and the thermostat is running on its built-in backup sensor only. Heating is limited to a safe minimum range and cooling is disabled.	See the troubleshooting section; if it persists, call your installer.

When to call a professional

Call a licensed HVAC technician (do not attempt yourself) if:

- A furnace fault code appears repeatedly.
- The “no heat — system in safe state” alert appears.
- You hear unusual noises, smell burning or gas, or see water around the furnace or indoor coil.
- Heating or cooling performance changes suddenly without a weather explanation.

Routine actions you **can** do yourself: changing setpoints and modes, replacing the furnace filter per the furnace manual, replacing remote-sensor batteries, and clearing snow or debris from around the outdoor unit.

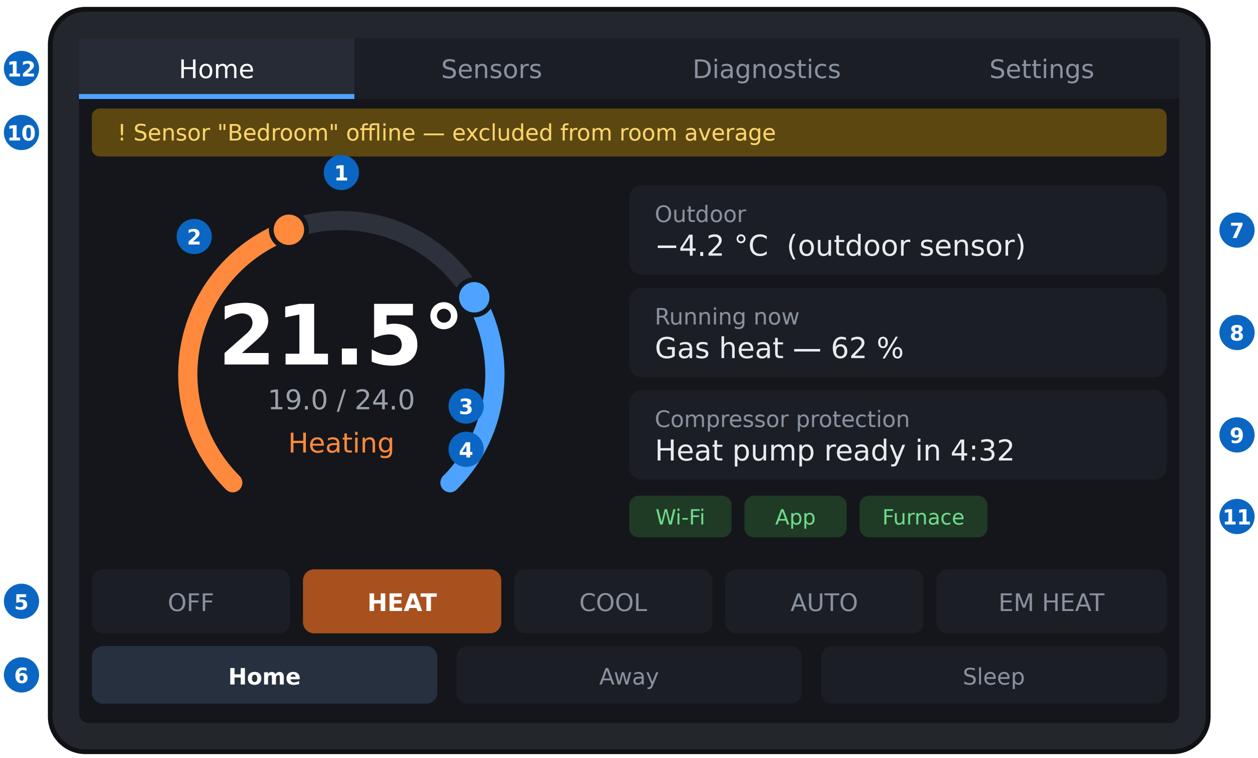
A note on what this thermostat will never do

- It never disables or bypasses any furnace or heat pump safety device — those are physically out of its reach by design.
- It never runs gas heat and heat pump heat together (briefly tempering the air during the heat pump’s defrost cycle is the one designed exception).
- It never restarts the compressor faster than the compressor’s protection timers allow — even while recovering from a fault.
- When in doubt, it stops requesting heating and cooling and tells you.

Getting to know your thermostat

The thermostat is a 4.3-inch colour touchscreen mounted on the wall where your old thermostat was. Everything you need day to day is on the **Home** screen; three more pages (Sensors, Diagnostics, Settings) are a tap away.

The home screen



DT-1 home screen (illustration; colours and exact layout may differ slightly)

DT-1 home screen with numbered callouts

#	Element	What it shows / does
1	**Current temperature**	The room temperature the thermostat is controlling to — a blend of your room sensors (see *Remote sensors*). Shows `--.-` if no valid reading is available.
2	**Setpoint dial**	Two coloured arcs around the dial: orange for the **heat** setpoint, blue for the **cool** setpoint. Drag an arc's knob to change that setpoint. The two setpoints always keep a minimum gap (see *Everyday use*).
3	**Setpoint readout**	The two setpoints as numbers, e.g. `19.0 / 24.0` (heat / cool).
4	**Activity**	What the system is doing right now: *Idle*, *Heating*, *Cooling*, *Fan*, or *Defrosting*.
5	**Mode row**	Tap to choose **OFF / HEAT / COOL / AUTO / EM HEAT**. The active mode is highlighted.
6	**Preset row**	Tap **Home / Away / Sleep** comfort presets (see *Schedules and presets*).
7	**Outdoor temperature**	The outside temperature the thermostat is using, with its source (heating equipment, outdoor sensor, or weather service).
8	**Running now**	Which equipment is active: gas heat (with its current output, 40–100 %), heat

#	Element	What it shows / does
		pump heat, cooling, fan, or defrost.
9	**Compressor protection countdown**	When the heat pump is resting between runs, a countdown shows when it may start again. Hidden when not relevant. This is normal protection, not a fault.
10	**Alert banner**	The most recent alert, in plain language. Hidden when there is nothing to report. Also shows the “backup sensor mode” notice when applicable.
11	**Connection icons**	Status of **Wi-Fi**, the link to your **Home Assistant app**, and the link to the **furnace**. Green = good. The thermostat keeps controlling temperature even when Wi-Fi or the app link is down.
12	**Page tabs**	Switch between Home, Sensors, Diagnostics, and Settings.

The other pages

- **Sensors** — a table of every room sensor: its name, temperature, whether the room is occupied, how fresh the reading is, whether it is currently included in the room average, and its health. Useful for checking a sensor battery or seeing why a room is being ignored.
- **Diagnostics** — the alert history plus connection health (Wi-Fi, app link, furnace link), the compressor countdown, and the furnace’s current output. This is the page a service technician will ask you to read out.

- **Settings** — mostly informational on the wall unit: the active minimum gap between heat and cool setpoints, and software version information. Detailed tuning (schedules, balance point, sensor options) is done in the Home Assistant app, not on the wall.

Things worth knowing

- **The screen is the boss’s messenger, not the boss.** Everything you tap is double-checked by the thermostat’s control core before any equipment is asked to run. Out-of-range requests are corrected automatically.
- **It works without the network.** Wi-Fi loss never stops heating or cooling. You will see the Wi-Fi icon turn grey; local control continues.
- **Temperatures are in °C** in 0.5° steps.

Everyday use

Setting the temperature

From the wall screen:

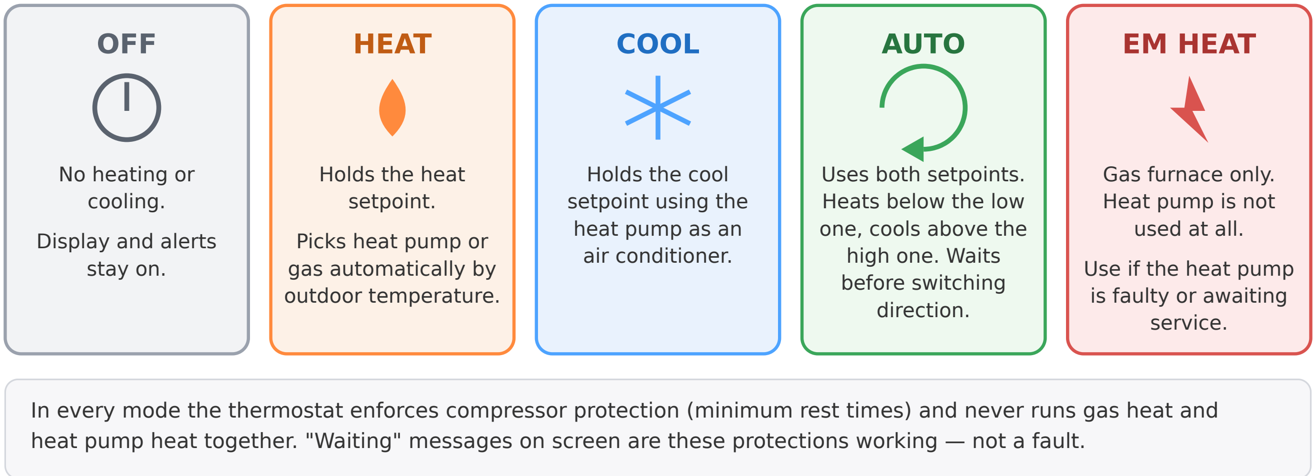
1. On the **Home** screen, find the dial with the orange (heat) and blue (cool) arcs.
2. Drag the **orange knob** to change the heat setpoint — the temperature below which heating starts.
3. Drag the **blue knob** to change the cool setpoint — the temperature above which cooling starts.
4. The new values appear immediately in the setpoint readout. There is nothing to “save”.

From your phone: open the thermostat card in the Home Assistant Companion app and drag the same setpoints there (see *Mobile app*).

The minimum gap. The cool setpoint must always be at least a set amount above the heat setpoint (factory default 2.8 °C). If you move one setpoint too close to the other, the thermostat automatically pushes the other one along to keep the gap. This prevents the system from heating and cooling the same air in turns.

Choosing a mode

Tap a button on the mode row: **OFF / HEAT / COOL / AUTO / EM HEAT**.



Modes at a glance

- **OFF** — no heating or cooling. The screen, sensors, and alerts stay active. Note: there is no “away for the winter” frost-protection in OFF — if you leave for an extended period in winter, use HEAT or AUTO with a low heat setpoint instead.
- **HEAT** — holds the heat setpoint. The thermostat automatically picks the heat source: the **heat pump** in milder weather, the **gas furnace** in colder weather (see “How the thermostat chooses” below). You don’t manage this — it is what “dual fuel” means.
- **COOL** — holds the cool setpoint, using the heat pump as a central air conditioner. To protect the equipment, cooling will not run when the indoor temperature is below 18 °C.
- **AUTO** — uses **both** setpoints and changes over between heating and cooling by itself. Heats when the room drops below the heat setpoint, cools when it rises above the cool setpoint, does nothing in between. Ideal for spring and fall, when mornings need heat and afternoons need cooling.
- **EM HEAT (emergency heat)** — gas furnace only; the heat pump is not used at all. Use this if the heat pump is damaged, iced up, or awaiting service. Emergency heat is selected on the wall screen; in the phone app the mode list shows Off / Heat / Cool / Auto.

Choosing a preset

Tap **Home**, **Away**, or **Sleep** on the preset row. Each preset is a pair of setpoints you (or your installer) define in Home Assistant — for example, Away might be heat 16 /

cool 28 to save energy while nobody is in. Schedules normally switch presets for you automatically (see *Schedules and presets*).

Why the thermostat sometimes waits

You will occasionally change a setting and hear… nothing. A short wait is almost always deliberate protection, not a problem:

- **Compressor rest time.** The heat pump's compressor must rest for about 5 minutes after it stops before it may start again, and once started it runs at least 5 minutes. Starting a compressor against pressure that hasn't equalized shortens its life dramatically — every quality thermostat enforces this. The home screen shows a countdown ("Heat pump ready in 4:32") while it applies. The thermostat also limits the compressor to 3 starts per hour and waits about 5 minutes after a power outage before the first compressor start.
- **Changeover wait (AUTO mode).** Switching between heating and cooling has its own patience built in: the system waits a sustained period (about 10 minutes of genuine need, and at least 30 minutes since the opposite operation) before reversing direction. This stops a sunny hour from flipping your system from heat to cool and back.
- **Reaction threshold.** The thermostat doesn't chase every 0.1° wiggle — it acts when the room moves meaningfully past a setpoint (about 0.5 °C).

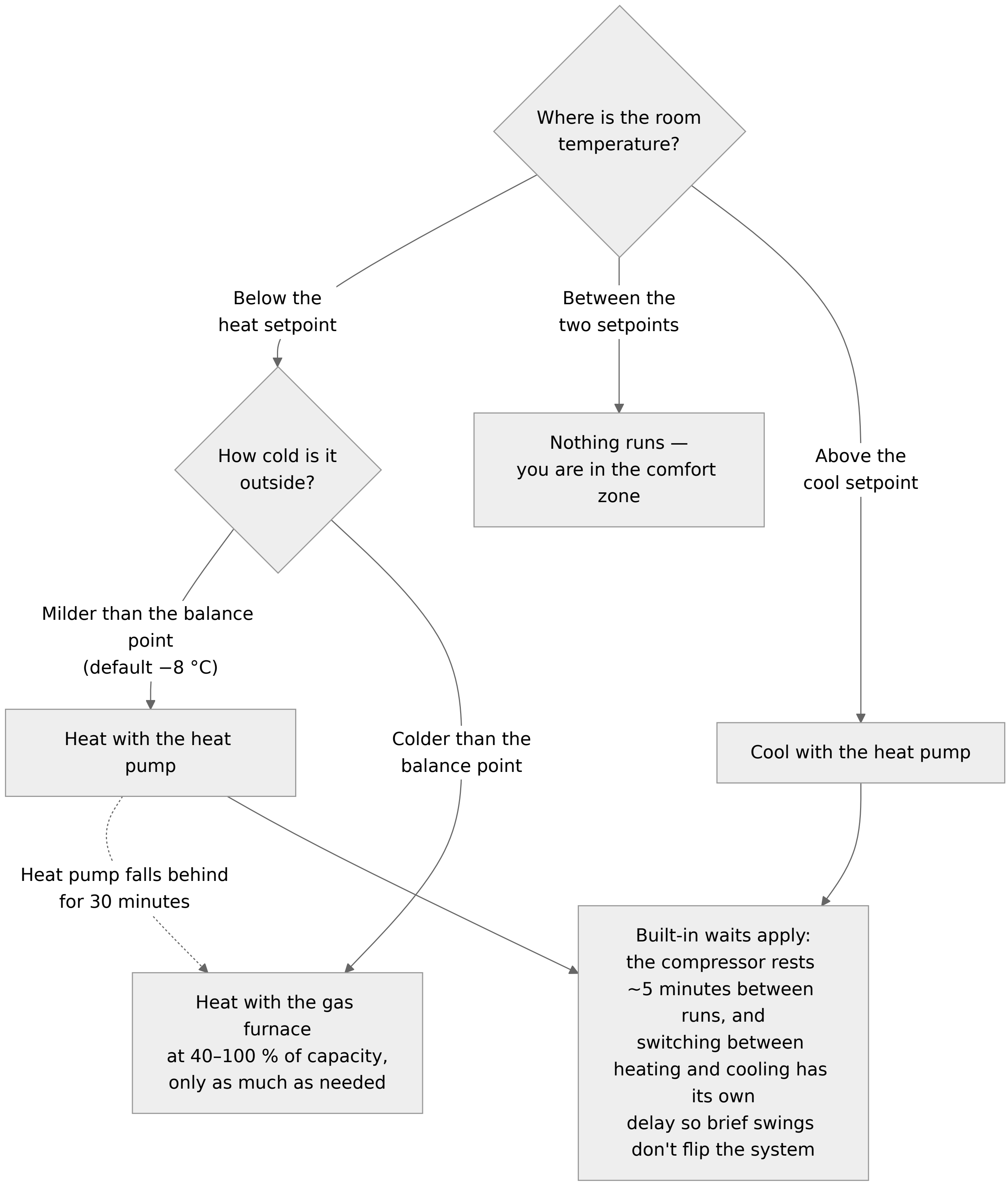
How the thermostat chooses heat pump vs gas

In HEAT and AUTO modes the choice is automatic, based mainly on outdoor temperature:

- **Milder than the balance point** (factory default −8 °C): the heat pump heats. It may run for long stretches near this temperature — for an inverter heat pump, long gentle runs are normal and efficient, not a fault.
- **Colder than the balance point**: the gas furnace heats, modulating smoothly between 40 % and 100 % of its capacity to match the need.
- **The heat pump falls behind**: if the room droops well below the setpoint for half an hour with the heat pump flat out, the thermostat steps over to gas until conditions improve.
- **Very cold weather**: below the compressor's safe operating limit (default −20 °C outdoor), the heat pump is locked out entirely and gas carries the load.

- **Mild weather**: above about 10 °C outdoor, gas heat is locked out and the heat pump handles any heating need.

Gas heat and heat pump heat never run at the same time — the equipment manufacturer prohibits it, and the thermostat enforces it. The one designed exception: during the heat pump's **defrost cycle** in winter, the furnace briefly tempers the air so the vents don't blow cold (see below).



How AUTO mode decides

About defrost (winter, heat pump running)

When the outdoor unit frosts up in cold, damp weather it periodically runs a defrost cycle: the outdoor fan stops, you may see steam rising from the outdoor unit and hear a whoosh or hiss, and the screen shows **Defrosting**. The system briefly adds a little furnace heat during the cycle so the indoor air stays comfortable. Defrost lasts a few minutes and is completely normal.

After a power outage

The thermostat restarts in a deliberately cautious state: it re-checks its sensors and settings before asking anything to run, and holds the heat pump off for about 5 minutes. Your modes, setpoints, and presets are remembered. If the outage interrupted a compressor run, expect the heat pump to wait out its full rest time before restarting.

Schedules and presets

How scheduling works — an honest explanation

The wall thermostat itself **does not store a weekly schedule**. Scheduling lives in **Home Assistant**, the home-automation system the thermostat connects to. Home Assistant changes the thermostat’s preset (or setpoints) at the times you choose, and the thermostat carries them out.

This split is deliberate:

- Home Assistant’s scheduler is far more capable than anything that would fit on a wall screen — different weekday/weekend programs, presence-based rules (“switch to Away when both phones leave”), calendar integration, and easy editing from your phone or computer.
- The thermostat stays simple and dependable: it always knows its current setpoints and keeps controlling temperature even if Home Assistant goes quiet.

What this means for you: to set up or change a schedule, you use the Home Assistant app or web page — not the wall screen. If you never set up Home Assistant, the thermostat simply holds whatever setpoints and mode you set by hand, like a classic non-programmable thermostat.

Presets: Home, Away, Sleep

A preset is a named pair of setpoints. The three presets are:

Preset	Typical use	Example setpoints (heat / cool)
Home	Normal occupied comfort	21 / 24 °C
Away	Nobody in for hours — save energy	16 / 28 °C
Sleep	Overnight	18 / 26 °C

The example values are only suggestions — the actual setpoints behind each preset are configured in Home Assistant to suit your household.

You can switch presets three ways:

1. **Automatically by schedule** — the usual way: e.g. Sleep at 22:30, Home at 06:30, Away on weekday work hours.
2. **By tapping the preset row** on the wall screen — handy when the schedule guessed wrong (“we’re home early”).
3. **From the phone app** — the preset selector on the thermostat card.

A manual preset or setpoint change simply stays in effect until the next scheduled change comes along.

Presets can also influence which room sensors are used — for example, Sleep can be configured to follow the bedroom sensors only. See *Remote sensors*.

A starter schedule

If you are setting up Home Assistant scheduling for the first time, this pattern serves most households well:

Time	Weekdays	Weekends
06:30	Home	Home (or 07:30)
08:30	Away (if the house empties)	—
17:00	Home	—
22:30	Sleep	Sleep

Set it up in Home Assistant with a Scheduler card/automation targeting the thermostat’s preset. Your installer may have already created one for you.

If Home Assistant goes down

Nothing dramatic happens. The thermostat keeps its last setpoints and mode and continues controlling. If it hears nothing from Home Assistant for more than 30 minutes, it falls back to a conservative safety net — heat to 18 °C, cool above 27 °C, in your last chosen mode — so the home can neither freeze nor swelter while the network is out. When Home Assistant returns, schedules resume on their own.

Remote room sensors

What they do

A thermostat that only measures the hallway keeps the *hallway* comfortable. Remote sensors let the thermostat see the rooms you actually live in:

- **Room averaging.** The thermostat blends readings from all healthy sensors into one “effective” room temperature and controls to that.
- **Follow-me comfort.** Sensors that detect occupancy get roughly double the influence of empty rooms, shifting comfort toward where people actually are. The shift is gradual (over tens of minutes), so walking through a room doesn’t yank the temperature around.
- **Per-preset rooms.** Presets can use different sensor groups — typically Sleep follows the bedrooms, Home follows the living areas.

The blending happens inside the thermostat itself, so an established sensor average keeps working even during a network or Home Assistant hiccup.

Supported sensors

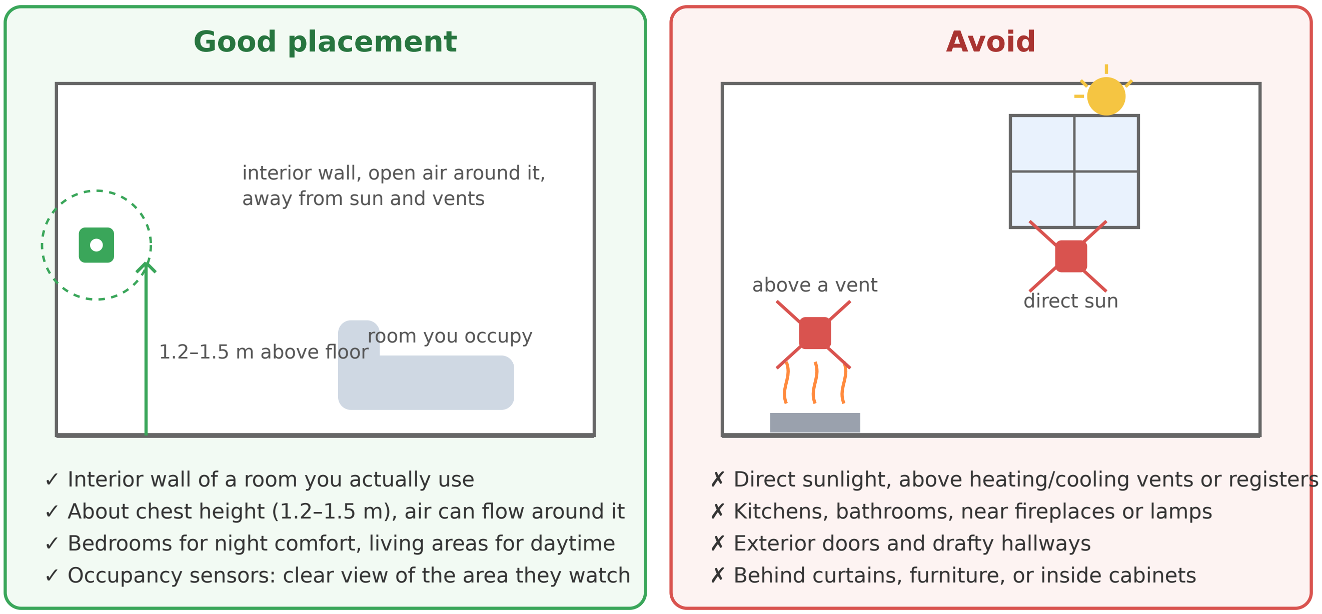
Sensors connect through Home Assistant — they are standard smart-home devices, not proprietary thermostat accessories:

Type	Examples	Notes
Zigbee temperature	SONOFF SNZB-02D, Aqara temperature sensors	Battery powered, small, cheap; needs a Zigbee radio on your Home Assistant box
Occupancy (mmWave)	Aqara FP2, Apollo MSR-2	Detects presence even when you sit still — better than simple motion sensors for follow-me
ESPHome-based	Apollo MSR-2 and other ESPHome devices	Wi-Fi devices managed from Home Assistant

Any temperature or occupancy sensor that Home Assistant can read can be bridged to the thermostat. Your installer sets up the bridge and the sensor roster. (Ecobee's own wireless sensors are **not** compatible — they use a proprietary radio protocol that only Ecobee thermostats can receive.)

The thermostat also has its own built-in wired backup sensor, which cross-checks the room average and takes over if every remote sensor fails.

Where to put sensors



Sensor placement do and don't

Do:

- Mount on an **interior wall** of a room you actually use, about **1.2–1.5 m above the floor** (chest height), with open air around it.
- Put one in the **main living area** and one in the **bedroom(s)** you want comfortable at night.
- Give occupancy sensors a clear view of the space they watch.

Avoid:

- Direct sunlight, and spots above or near **heating/cooling vents, radiators, lamps, or electronics** — these read the heat source, not the room.
- **Kitchens and bathrooms** (cooking and showers swing the reading).

- Exterior doors, drafty hallways, and uninsulated exterior walls.
- Behind curtains or furniture, or inside cabinets.

What happens when a sensor misbehaves

The thermostat checks every sensor continuously and protects itself from bad data:

- **Sensor goes quiet** (dead battery, out of range): after about 5 minutes without a fresh reading, the sensor is dropped from the average and listed as stale on the Sensors page. The remaining sensors carry on.
- **Sensor reads nonsense** (frozen value, impossible temperature, or more than 4 °C away from what the other sensors agree on): it is excluded and an alert names the sensor.
- **A sensor joins or leaves the average**: the effective temperature glides to the new value rather than jumping, so you won't get a sudden burst of heating or cooling because a battery died.
- **All remote sensors lost**: the thermostat falls back to its built-in backup sensor and enters **backup sensor mode** — heating is limited to a modest range (it keeps the home safe, around 16–18 °C, rather than precisely comfortable), **cooling is disabled**, and a persistent alert reminds you to fix the sensors. This is intentional: the backup sensor sits near the equipment, not in your living space, so the thermostat refuses to chase comfort with it.
- **Even the backup sensor fails**: the thermostat stops all heating and cooling requests and raises an alert — it will never run equipment on data it can't trust.

Routine care: replace sensor batteries when the Sensors page (or Home Assistant) shows them low — typically once a year or two for Zigbee sensors.

Mobile app and remote control

The short version

There is no “ElectricRV app” to download — and that’s a feature. Phone control uses the free **Home Assistant Companion app** (iOS and Android), talking to *your own* Home Assistant system inside your own home. No cloud account, no subscription, no company server between your finger and your furnace.

From the app you can:

- See the current temperature and adjust both setpoints.
- Change modes (Off / Heat / Cool / Auto) and presets (Home / Away / Sleep).
- See what’s running (heating, cooling, defrosting) and the outdoor temperature.
- Edit schedules, view temperature history graphs, and receive alert notifications.

Setting it up

Your installer normally completes this at commissioning. If you’re adding a phone later:

1. Install **Home Assistant Companion** from the App Store or Google Play.
2. Open the app **while connected to your home Wi-Fi**. It discovers your Home Assistant system automatically (or enter its address, e.g. `http://homeassistant.local:8123`).
3. Sign in with the Home Assistant user account your installer created for your household (each family member can have their own).
4. Open the **thermostat card** — your installer will have placed it on the main dashboard. Tap the star to favourite it if you like.

That’s all — on your home Wi-Fi, control is instant and fully local.

Control from outside the home

Because there’s no cloud middleman, reaching the thermostat from outside your house requires a secure tunnel into your home network. Two good options, both typically set up by your installer:

- **Tailscale or WireGuard (recommended, free).** A private, encrypted “virtual cable” between your phone and your home network. Once installed on the phone, the Companion app works from anywhere exactly as it does at home. Tailscale in particular is a simple app install plus account — no router surgery.
- **Home Assistant Cloud (Nabu Casa, paid subscription).** A zero-setup relay service from the Home Assistant developers (~US$6.50/month). It is the easy button, but it does route through a third-party server, which runs against the local-only philosophy of this product. Your choice.

Privacy note: with the recommended setup, your temperature data, occupancy data, and control of your furnace never leave your home network.

If the app can’t connect

The thermostat does not need the app to function — the wall screen always works, and temperature control continues uninterrupted. App connection problems are network problems, in this order of likelihood:

1. Your phone is not on the home Wi-Fi (or the VPN/Tailscale is off, if you’re away from home).
2. The Home Assistant computer is off or restarting.
3. Your router or Wi-Fi is down.

If Home Assistant stays unreachable for more than 30 minutes, the wall thermostat quietly switches to its built-in fallback comfort range (heat to 18 °C / cool above 27 °C) until contact resumes — see *Schedules and presets*.

Alerts and troubleshooting

The thermostat reports problems in the alert banner on the home screen, on the Diagnostics page, and (when connected) as notifications in the Home Assistant app. This section explains what each message means and what to do.

> **First, the golden rule:** when this thermostat is unsure, it *stops asking for heating and cooling* and tells you. Most alerts therefore mean “comfort may suffer until this is fixed”, not “something dangerous is happening”. The dangerous-condition protections live inside the furnace and heat pump themselves and are always active.

“Waiting” states (normal — not faults)

What you see	What it means	What to do
“Heat pump ready in M:SS” countdown	The compressor is resting between runs (about 5 minutes), or the thermostat is limiting it to 3 starts per hour, or the system just powered up.	Nothing — it starts by itself when the countdown ends.
Changed AUTO direction but nothing happens	Switching between heating and cooling requires the need to persist about 10 minutes, and at least 30 minutes since the opposite operation.	Wait. If you need cooling *right now*, switch the mode to COOL manually.
“Defrosting” with steam/whoosh at the outdoor unit	Normal winter defrost of the heat pump; the furnace briefly tempers the air.	Nothing — a few minutes, then normal operation resumes.
Heat pump runs for hours without stopping	Normal for an inverter heat pump near its balance point — long, low-power runs are its most efficient behaviour.	Nothing, if the room is holding temperature.
Cooling refuses to start in a cold room	Cooling is locked out when the indoor temperature is below 18 °C, to protect the equipment.	Nothing — this is by design.

Alert table

Alert / symptom	What it means	What to do
Sensor “name” offline / excluded	That room sensor stopped reporting (likely a dead battery) or disagreed wildly with the others. The system continues on the remaining sensors.	Check the sensor’s battery and placement; see the Sensors page for details.
Backup sensor mode (persistent banner)	*All* remote sensors are unavailable; the thermostat is running on its built-in backup sensor. Heating is limited to a safe modest range (~16–18 °C) and **cooling is disabled**.	Restore the sensors: check sensor batteries, then whether Home Assistant and your network are running. If it persists, call your installer.
Heat pump locked out — outdoor temperature	It is colder outside than the compressor’s safe limit (default −20 °C). Gas heat carries the load automatically.	Nothing — normal cold-weather behaviour.
Heating switched to gas — heat pump couldn’t keep up	The heat pump ran flat out but the room kept drooping, so gas took over.	Nothing — this is the dual-fuel design working. Frequent occurrences in mild weather are worth mentioning to your installer.
Gas heating stopped — ran 4 hours without progress	A safety limit: gas heat ran continuously for 4 hours without gaining on the setpoint, so the thermostat dropped the call and alerted you.	Look for open windows/doors, a clogged filter, or an extreme-weather day. Clear/acknowledge and let it retry; if it recurs, call a professional.
Furnace fault (code …)	The furnace’s own control board reported a problem. The thermostat displays it but cannot fix or override it.	Note the code from the Diagnostics page and call your HVAC service company.

Alert / symptom	What it means	What to do
Wi-Fi / App icon grey	Network or Home Assistant connection lost. Temperature control continues locally; after 30 minutes the fallback comfort range (heat 18 / cool 27 °C) applies until reconnection.	Check your router and the Home Assistant computer. The wall screen keeps full control meanwhile.
Furnace link lost	The thermostat cannot talk to the furnace over its communication wiring. Heating requests cannot be delivered.	Power-cycling your router won't help — this is the wired link. If it doesn't clear within minutes, call your installer. In winter treat this as urgent (heat is unavailable).
No heat — system in safe state (escalating)	The thermostat hit a fault it could not resolve and stopped all requests. The alert repeats and escalates because an unheated home in winter risks frozen pipes.	Call your installer or HVAC service promptly. Your installer prepared a backup-thermostat changeover for exactly this case.
System restarted repeatedly — locked safe	The controller restarted several times in a short period and locked itself in the no-demand state until manually cleared.	Call your installer. Do not repeatedly power-cycle it yourself.
Blank screen	The thermostat has no power. It is powered from the furnace, so the furnace likely has no power either.	Check the furnace's switch and breaker. If power is on and the screen stays dark, call your installer.
Room comfortable but a far room is cold/hot	The thermostat controls to its sensor average; rooms without sensors (or with closed doors/vents) drift.	Add a sensor to that room, or check vents and doors.
Vents blow cool-ish air in heat-pump heating	Heat pump supply air (often 30–40 °C) feels cooler than gas-furnace air, especially	Nothing, if the room holds temperature.

Alert / symptom	What it means	What to do
	on your hand — but it is still heating.	

Reading the Diagnostics page to a technician

When you call for service, open **Diagnostics** and have ready:

1. The current alerts (and any furnace fault code).
2. The connection health rows (Wi-Fi / App / Furnace).
3. What was running at the time ("Running now" on the home screen).

> ⚠ **To be confirmed during installation:** the exact wording of furnace fault codes shown on the Diagnostics page depends on the installed equipment configuration and will be finalized after on-site verification. Treat any fault code as "call for service" until this manual is updated.

Specifications

Wall unit

Item	Specification
Display	4.3-inch colour IPS touchscreen, 480 × 272, capacitive touch, landscape
Power source	24 V AC from the furnace (no batteries; no wall adapter)
Power consumption	≤ ~3 VA from the furnace transformer (about 2 W; measured for this class of display board, to be confirmed on the production unit)
Connectivity	Wi-Fi (2.4 GHz) to your home network; wired digital link to the furnace
Smart-home integration	Home Assistant (local network, MQTT); Home Assistant Companion app
Built-in backup sensor	Wired digital temperature sensor
Mounting	Replaces the existing thermostat at the same wall location (professional installation)

Control ranges and behaviour

Item	Value
Setpoint range	10–30 °C (the wall dial may display a wider range, but every request is validated by the controller; 10–30 °C is the effective control range)
Setpoint adjustment step	0.5 °C
Minimum heat/cool setpoint gap	2.8 °C default; never below 1.1 °C
Reaction threshold (hysteresis)	~0.55 °C past setpoint
Gas furnace output range	0 (off), or 40–100 % of capacity (modulating)
Cooling lockout	No cooling when indoor temperature is below 18 °C
Remote sensor accepted range	5–40 °C (readings outside are rejected as faulty)
Remote sensor timeout	Dropped from the average after ~5 min without a fresh reading

Protection timers and dual-fuel defaults

These are factory defaults; your installer can tune them within safe bounds.

Parameter	Default
Compressor minimum off time	5 min
Compressor minimum run time	5 min
Compressor starts per hour (max)	3
Compressor hold-off after power-up	~5 min
Heat/cool changeover: sustained need required	10 min
Heat/cool changeover: minimum time since opposite call	30 min
Dual-fuel balance point (heat pump vs gas)	−8 °C outdoor (±2 °C switching band)
Compressor low-temperature lockout	−20 °C outdoor
Gas/aux heat lockout (mild weather)	above 10 °C outdoor
Heat-pump-to-gas escalation	room ≥1 °C below setpoint for 30 min at full heat-pump output
Gas heat maximum continuous run before alert/stop	4 h
Defrost air tempering	a small amount of furnace heat during defrost (exact behaviour finalized during installation)
Network-loss fallback setpoints (after 30 min)	heat 18 °C / cool 27 °C, last mode kept
Backup-sensor-mode heating range	16–18 °C (cooling disabled)

> ⚠ **To be confirmed during installation:** heat pump operating limits (including the exact low-temperature rating of the installed outdoor unit) are confirmed against the installed model during commissioning; the −20 °C lockout default is set conservatively until then.

Compatible equipment

Role	Equipment
Gas furnace	Dettson Chinook modulating gas furnace (communicating control)
Heat pump / air conditioning	Gree-built heat pump matched to the Chinook
Remote sensors	Zigbee and ESPHome temperature/occupancy sensors via Home Assistant (e.g. SONOFF SNZB-02D, Aqara, Apollo MSR-2, Aqara FP2)

This thermostat is purpose-built for the equipment combination above. It is not a general-purpose replacement thermostat for other furnaces or heat pumps.

Glossary

Auto-changeover (AUTO mode) — the thermostat switches between heating and cooling by itself, using both setpoints, with built-in waits so brief temperature swings don’t flip the system back and forth.

Backup sensor mode — a degraded-but-safe state entered when all remote room sensors are unavailable: the thermostat runs on its built-in sensor, limits heating to a modest range, disables cooling, and alerts you.

Balance point — the outdoor temperature (default −8 °C) below which the thermostat prefers gas heat over the heat pump.

Compressor — the heart of the heat pump; the motor-driven pump that moves refrigerant. It needs rest periods between starts, which is why the thermostat sometimes shows a countdown before the heat pump runs.

Defrost — a normal winter cycle in which the heat pump briefly melts frost off its outdoor coil. Steam from the outdoor unit and a whooshing sound are normal; the furnace tempers the indoor air meanwhile.

Dual fuel — a system with two heat sources (here: heat pump and gas furnace) where a controller picks the better one for the conditions.

Emergency heat (EM HEAT) — a mode that uses the gas furnace only, bypassing the heat pump entirely. For when the heat pump is faulty or being serviced.

ESPHome — an open-source system for Wi-Fi smart-home sensors and devices that integrate with Home Assistant. Some supported room sensors are ESPHome devices.

Follow-me — comfort weighting that gives occupied rooms more influence on the controlled temperature, using occupancy-capable sensors.

Heat pump — equipment that heats or cools by moving heat between indoors and outdoors. Very efficient in mild weather; less effective in severe cold, which is why this system pairs it with a gas furnace.

Home Assistant — free, open-source home-automation software that runs on a small computer in your home. Provides the thermostat’s schedules, phone app, sensor bridge, and history — all locally, with no cloud.

Modulating furnace — a furnace that can run anywhere between 40 % and 100 % of its capacity instead of just on/off, giving longer, quieter, more even heating. This thermostat requests the exact firing rate (40–100 %) and the furnace's certified control carries it out.

Occupancy sensor — a sensor that detects whether people are present in a room. mmWave types (radar-based) detect even motionless people.

Preset — a named pair of setpoints (Home / Away / Sleep) switched by schedule, by tap, or from the app.

Setpoint — a target temperature. This thermostat keeps two: the heat setpoint (heating starts below it) and the cool setpoint (cooling starts above it).

Short-cycle protection — the timers that prevent the compressor from restarting too soon or too often, protecting it from damage.

Zigbee — a low-power wireless standard used by many battery-powered smart-home sensors; connects to Home Assistant through a Zigbee radio.